

Singular Value Decomposition (SVD)

Definition

Singular Value Decomposition (SVD) is a method in linear algebra used to factorize any matrix into three simpler matrices.

For any matrix A ,

$$A = U\Sigma V^T$$

Where:

- U → Orthogonal matrix containing left singular vectors
 - Σ (**Sigma**) → Diagonal matrix containing singular values
 - V^T → Transpose of orthogonal matrix containing right singular vectors
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Components of SVD

1. Matrix U

- Size: $m \times m$
- Contains left singular vectors
- Orthogonal matrix

Properties:

$$U^T U = I$$

2. Sigma Matrix (Σ)

- Diagonal matrix
- Contains singular values

$$\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq \dots \geq 0$$

- Singular values are always non-negative
 - They represent stretching factors
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3. Matrix V^T

- Transpose of matrix V
- Contains right singular vectors
- Orthogonal matrix

Properties:

$$V^T V = I$$

Geometric Interpretation

SVD performs three operations:

1. V^T rotates the input space
2. Σ stretches or compresses the space
3. U rotates the transformed space

Thus:

Rotation $\rightarrow$ Scaling $\rightarrow$ Rotation

Singular Values

The diagonal elements of Σ are called singular values.

- Large singular values $\rightarrow$ important information
 - Small singular values $\rightarrow$ less important information or noise
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Applications of SVD

1. Data Compression

Used to reduce storage while preserving important information.

Example:

- Image compression
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2. Noise Reduction

Small singular values often represent noise. Removing them helps clean the data.

3. Dimensionality Reduction

SVD is closely related to Principal Component Analysis (PCA).

Used in:

- Machine Learning
 - Data Science
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4. Solving Linear Systems

Useful for solving systems when matrices are singular or nearly singular.

Visual Intuition

If a matrix transforms a circle into an ellipse:

- Directions of ellipse axes → singular vectors
 - Lengths of ellipse axes → singular values
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Important Properties

- Works for square and rectangular matrices
 - Singular values are always non-negative
 - Gives optimal low-rank approximation
 - Numerically stable
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Summary

SVD decomposes a matrix into:

$$A = U\Sigma V^T$$

Where:

- U and V provide directions
- Σ provides scaling information

SVD is widely used in:

- Machine Learning
- PCA
- Image Compression
- Signal Processing
- Recommendation Systems
- Data Analysis